# Micro Project 1: Campus City Emergency Distribution

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Github Repository: `https://github.com/Aman-Govind/campus-emergency-supply-optimization.git`

## 1 Problem Statement

Campus City Logistics currently operates an inefficient ad-hoc system for distributing essential supplies to multiple campus facilities. The objective of this project is to design an optimized distribution network that ensures all facilities receive their required supplies while minimizing the total annual cost.

The system must select exactly two warehouses and determine the quantity of supplies transported to each facility. The total cost includes transportation cost, warehouse construction cost (amortized over ten years), and operational cost. The solution must also satisfy warehouse capacity limits and remain within the allocated annual budget of $1.5 million.

## 2 Mathematical Formulation

The problem is modeled as a Mixed Integer Linear Programming (MILP) problem.

### Decision Variables

$x_{w,f}$ : Annual units shipped from warehouse $w$ to facility $f$.

$y_w$ : Binary variable indicating whether warehouse $w$ is selected.

### Objective Function

Minimize the total cost:

$$Min\ Z = \sum (T_{w,f} \cdot x_{w,f}) + \sum \left( \frac{Const_w}{10} + Op_w \cdot 365 \right) y_w$$

### Constraints

Demand constraint ensures that every facility receives its required annual demand.

Warehouse capacity constraint ensures that shipments from a warehouse do not exceed its maximum capacity.

Exactly two warehouses must be selected to maintain redundancy in the system.

The total annual cost must remain within the budget limit of $1,500,000.

# 3 Methodology

The optimization model was implemented using Python with the PuLP library, which is commonly used for solving linear and mixed integer programming problems. The datasets containing facility information, warehouse data, transportation costs, and demand values were processed using the Pandas library.

The MILP model was constructed by defining decision variables, objective functions, and constraints. The CBC solver in PuLP was then used to compute the optimal warehouse selection and shipment plan. Geographic visualization of the logistics network was created using the Folium library to display warehouse locations, facilities, and transportation routes.

# 4 Results & Interpretation

The optimization model identified **WH_SOUTH** and **WH_WEST** as the optimal warehouses for supplying campus facilities. These locations were selected because they provide the lowest overall transportation and operational cost while satisfying all demand and capacity constraints.

The optimal shipment plan ensures that every facility receives its required annual supply. Facilities are served by the nearest warehouse whenever possible, which helps reduce transportation cost and improves distribution efficiency. High-demand locations such as the medical center and dormitories receive larger shipment volumes from the selected warehouses.

The total optimized annual cost obtained from the model is **$1,150,263**. This includes transportation cost, warehouse construction cost (amortized over ten years), and operational cost. The result confirms that the distribution network can meet all facility demands while remaining within the $1.5 million budget.

A snippet of the optimal shipment plan is shown in Table 1, which highlights how supplies are distributed from the selected warehouses to major campus facilities.

| Source Warehouse | Target Facility | Annual Units |
|---|---|---|
| WH_SOUTH | LOCAL_HOSPITAL | 29,565 |
| WH_WEST | MED_CENTER | 29,565 |
| WH_WEST | DORM_A | 17,885 |
| WH_SOUTH | DORM_B | 18,615 |
| WH_SOUTH | SCIENCE_HALL | 10,220 |

Table 1: Optimal Shipment Plan Snippet

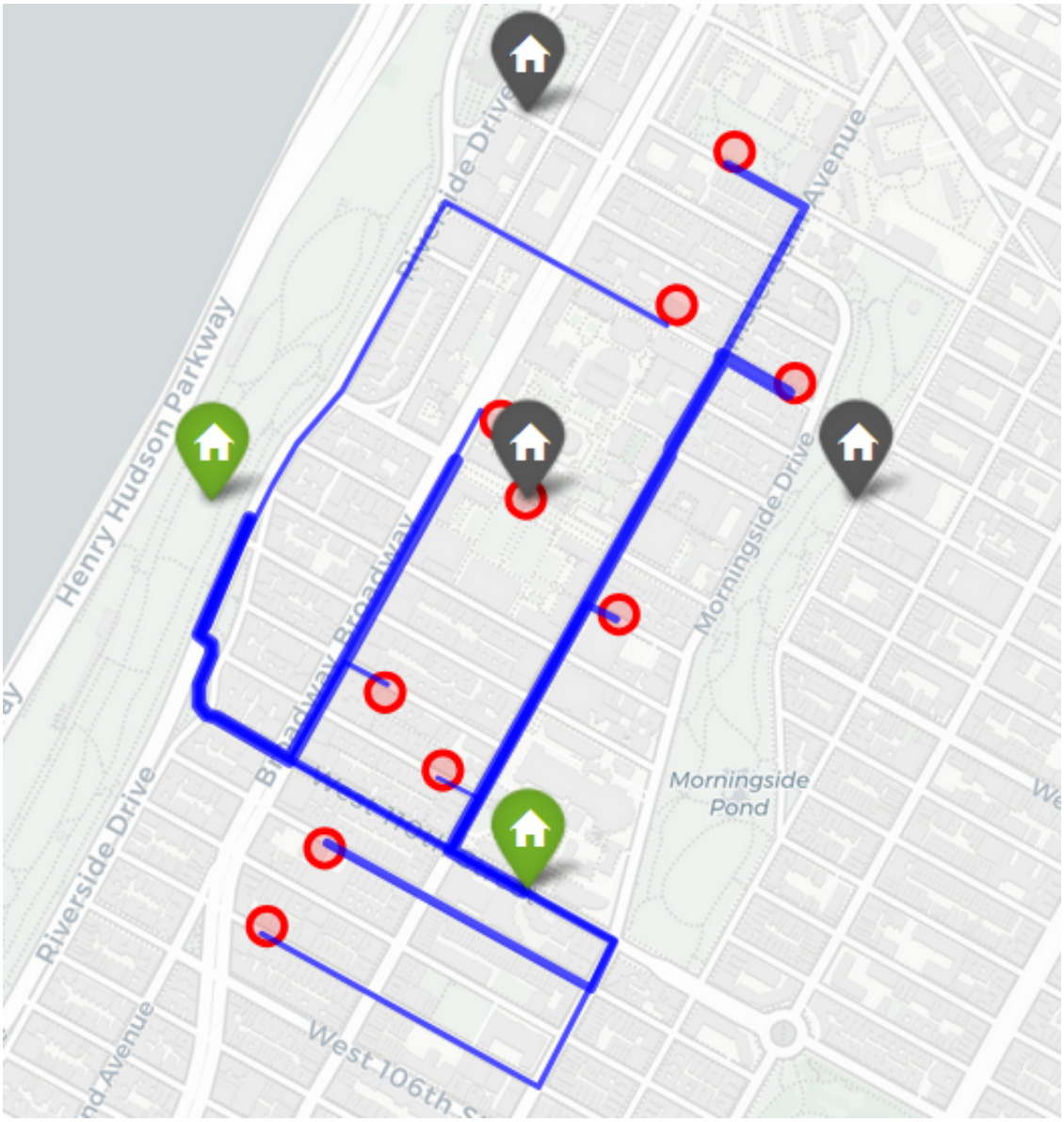


Figure 1: Campus Distribution Network

# 5 Conclusion

The proposed optimization model provides an efficient distribution strategy for emergency supplies across campus facilities. By applying Mixed Integer Linear Programming, the system determines the most suitable warehouse locations and shipment plan while minimizing total operational cost.

The results demonstrate that the optimized network can meet facility demands within the available budget while maintaining reliability in supply distribution.